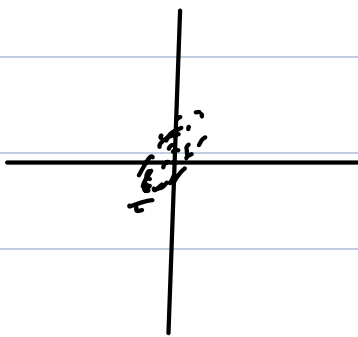


• Expectation
 $E[X]$

• Variance
 $E[(X - E[X])^2] = \text{Var}(X)$

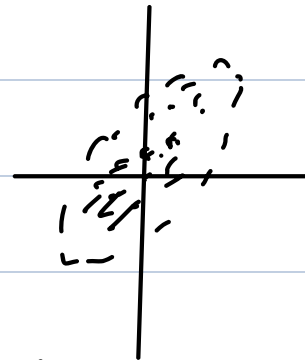
• Covariance
 $E[(X - E[X])(Y - E[Y])] = \text{Cov}(X, Y)$

Fact: Cov depends on the scale of the RVs.

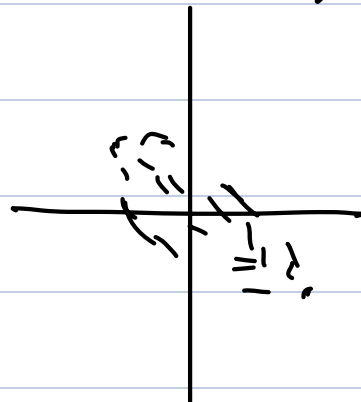
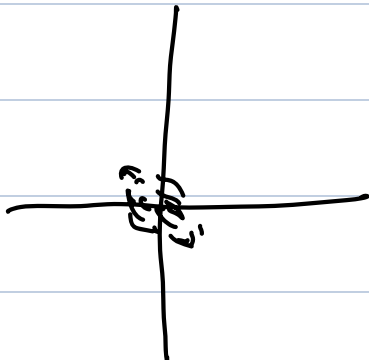


$|\text{Cov}(\dots)|$

$<$



$|\text{Cov}(\dots)|$



- Correlation (coefficient)

$$\rho(X, Y) = \text{Corr}(X, Y)$$

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

Assume
($\text{Var}(X) > 0$
 $\text{Var}(Y) > 0$)

- Fact: ρ does NOT depend on scale.
Precisely, a, b : 2 non-random constants

$$\rho(aX, bY) = \rho(X, Y)$$

- $-1 \leq \rho(X, Y) \leq 1$

- $\rho(X, Y) = 1 \Leftrightarrow Y = bX + a$
for some positive
slope $b > 0$

$$\rho(X, Y) = -1 \Leftrightarrow Y = bX + a$$

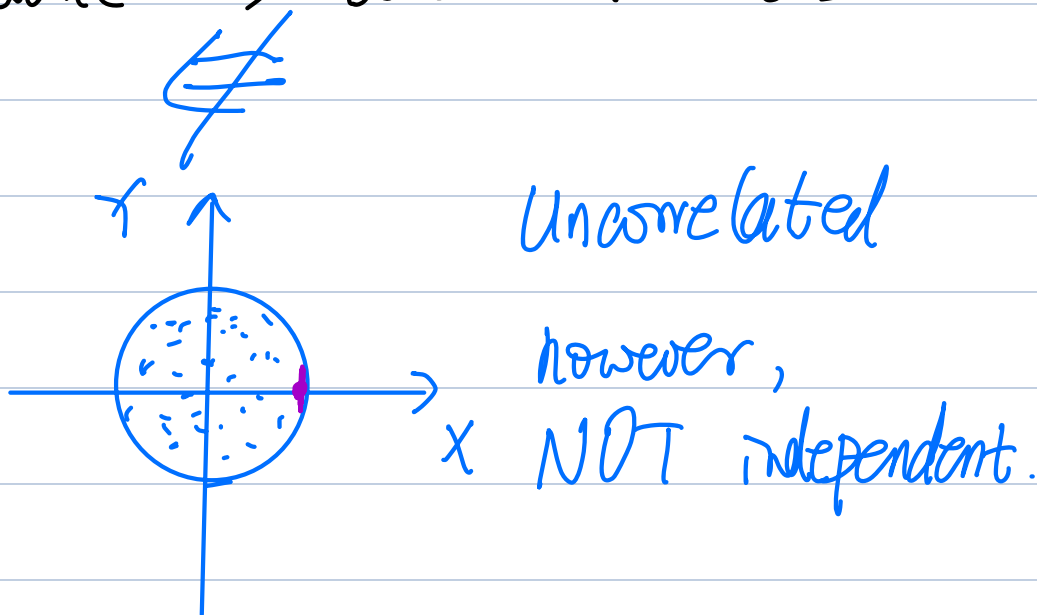
for some negative
slope $b < 0$.

- We say 2 RVs X & Y are

uncorrelated if $\rho(X, Y) = \text{Cov}(X, Y) = 0$

say 2 RVs X & Y are
independent if $P(X \in A | Y \in B)$
 $= P(X \in A)$ for any intervals A & B .
 $\Leftrightarrow P(X \in A, Y \in B)$
 $= P(X \in A) \cdot P(Y \in B)$

Independence $\Rightarrow$ Uncorrelatedness



Def. A time series (model) $\{w_t\}$ is said to be white noise, if

- $E[w_t] = 0$, for all t
- $\text{Var}[w_t] = \sigma_w^2 \in (0, \infty)$, for all t

- $\text{Cov}(w_s, w_t) = \text{Corr}(w_s, w_t) = 0$
for $s \neq t$.

$$V_t = \frac{1}{3} (w_{t-1} + w_t + w_{t+1})$$

$$V_{t+1} = \frac{1}{3} (w_t + w_{t+1} + w_{t+2})$$

Writing 3-term MA asy linear
filter formula.

$$Y_t = \sum_i a_{t-i} X_i$$

$$\underline{a_{-1} = a_0 = a_1 = \frac{1}{3}}, \quad a_i = 0 \text{ otherwise}$$

$$\begin{array}{ccc} t-i=-1 & t-i=0 & t-i=1 \\ \updownarrow & \updownarrow & \updownarrow \\ i=t+1 & i=t & i=t-1 \end{array}$$

$$\rightarrow Y_t = \sum_{t-1 \leq i \leq t+1} \frac{1}{3} X_i = \frac{1}{3} (X_{t-1} + X_t + X_{t+1})$$